Fabius Function—Missing Parts

The totalized inverse and the hidden machinery of the all-orders endpoint expansion

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Vladimir Reshetnikov's `ProveIt` repository

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Abstract

The Fabius function is unusually rich: a single compactly supported C^∞ density produces a self-similar distribution function, exact dyadic arithmetic, Fourier products, a probability model, and a small-argument expansion whose coefficients are smooth periodic functions. The companion article *The Fabius Function and Rvachev's Up-Function* develops most of that landscape. The present paper is deliberately different. It does not repeat the introductory theory. Instead it fills two identifiable gaps in the exposition by turning recently formalized material into ordinary mathematics.

The first gap is the inverse. We construct the totalized inverse $G = F^{-1}$ on the whole real line, prove its global monotonicity, continuity, reflection law, comparison calculus, and exact values at all outputs of the dyadic evaluator, and show how every flatness estimate for F becomes an effective steepness estimate for G. We then derive the classical interior calculus of the inverse and invert the sharp Lambert-phase asymptotic. The result is the explicit endpoint law

$$G(y) \sim \frac{\sqrt{\log(1/y)}}{\mathrm{e}\sqrt{\log 2}} \exp\Bigl(-\sqrt{2 \log 2 \, \log(1/y)}\Bigr) \qquad (y \downarrow 0),$$

together with its reflected analogue at 1. This explains in one formula why the inverse is steeper than every positive power of y, although it still tends to zero faster than every negative power of $\log(1/y)$.

The second gap concerns the internal machinery of the all-orders saddle expansion. We give complete derivations of the closed periodic jet formula, its shifted signed-Stirling conversion, the collapse of the centered exponent coefficients, the formal mass-to-logarithm recursions, the leading-derivative law, and the full second logarithmic correction. Finally, we give a proof-level account of the arbitrary-order analytic remainder: the order-dependent central window, vertical Taylor control, finite exponential substitution, Gaussian contraction, polynomial-tail replacement, minor-arc estimates, logarithmic transfer, and return from the dyadic logarithmic scale to $x \downarrow 0$.

Statements explicitly represented by theorem declarations in the Lean development are marked Lean. Consequences proved here from those statements by standard real or analytic arguments are marked Derived.

Contents

1 Purpose, scope, and status conventions

1.1 A supplement rather than a second survey

Let F denote the bounded, totalized Fabius function: $F(x) = 0$ for $x \le 0$, $F(x) = 1$ for $x \ge 1$, and on $[0, 1]$ it is the unique strictly increasing C^∞ function associated with Rvachev's compactly supported up-function. We use only a small portion of the foundational theory:

$$F(0) = 0, \qquad F(1) = 1, \qquad F(1-x) = 1 - F(x), \tag{1}$$

$$F'(x) = 2\,\mathrm{up}(2x-1), \qquad 0 < x < 1, \tag{2}$$

where up is positive on $(-1, 1)$ and $\mathrm{up}(0) = 1$. Consequently F is a continuous strictly increasing bijection of $[0, 1]$ onto itself.

The main companion article already explains the random-series model, infinite convolutions, the Fourier transform, moment recurrences, Thue–Morse identities, dyadic rationality, and the broad shape of the Lambert saddle analysis. Repeating those chapters would obscure the missing material. The division of labor adopted here is summarized in table 1.

Table 1: Expository audit and the role of this companion.

Topic	Situation in the main article	Treatment here
Inverse function	The totalized inverse module and its endpoint estimates are absent.	Full construction, exact identities, dyadic inversion, effective steepness, endpoint limits, regularity, and inverse asymptotics.
Closed saddle jets	Closed formulas are displayed, but the operator calculus and the shifted Stirling conversion are compressed.	Complete induction, forcing-term propagation, coefficient table, and the reason for the index shift $s(n+1, m+1)$.
Mass and logarithmic coefficients	The all-orders result is stated and some coefficients are listed.	Formal exponential recursion, Gaussian contraction, composition formulas, structural bounds, and the leading derivative law.
Second correction	The final polynomial appears, while much of the order-four algebra is easy to miss.	A line-by-line calculation from $E_1, \ldots, E_4$ through B_4, Gaussian moments, a_2, and A_2.
Arbitrary-order remainder	The main article gives a high-level proof architecture.	The order-dependent contour window and every analytic error mechanism are separated and justified.
Probability, Fourier theory, dyadic arithmetic	Already developed in substantial depth.	Used as input only; not duplicated.

1.2 What the labels mean

A paragraph headed Lean restates a mathematical theorem whose content is represented directly in the Lean files under

`Analysis/FabiusFunction/Lean/FabiusFunction`.

A paragraph headed DERIVED gives a paper-level consequence. Its proof uses formalized results plus a standard theorem such as the inverse function theorem, ordinary asymptotic inversion, or elementary formal power-series algebra. The distinction matters especially in sections 5 and 9: those sections contain new synthesis, but they do not pretend that every displayed corollary has a declaration of its own.

The exact Lambert phase is never silently replaced inside a periodic coefficient. If $\lambda = \lambda(x)$ is the exact lower solution of $x = \lambda 2^{-\lambda}$, then a term such as $A_j(\lambda)$ remains at that exact phase. Replacing λ by an asymptotic approximation inside A_j is a separate composition problem, because a bounded phase error does not become small modulo the period.

2 The inverse as a global mathematical object

2.1 The order isomorphism on the unit interval

Set

$$I = [0,1], \qquad \mathrm{clamp}_{[0,1]}(y) = \min\{1, \max\{0, y\}\}.$$

Because $F|_I$ is a continuous strictly increasing bijection $I \to I$, it is an order isomorphism. Its ordinary inverse will first be denoted by $F_I^{-1} : I \to I$.

Definition 2.1 (Totalized inverse)**.** The totalized inverse of the Fabius function is

$$G(y) = F_I^{-1}\big(\mathrm{clamp}_{[0,1]}(y)\big), \qquad y \in \mathbb{R}.$$

Thus the inverse is extended by constants outside its natural range, exactly as F itself is extended by constants outside its natural domain.

Theorem 2.2 (Global inverse package; LEAN)**.** *The function G has the following properties.*

1. $0 \le G(y) \le 1$ *for every* $y \in \mathbb{R}$.
2. *If* $0 \le y \le 1$, *then*
$$F(G(y)) = y.$$
If $0 \le x \le 1$, *then*
$$G(F(x)) = x.$$
3. *G is strictly increasing on* $[0,1]$, *monotone on* $\mathbb{R}$, *and continuous on* $\mathbb{R}$.
4. *The restriction $G|_I$ is a bijection* $I \to I$.

Proof. The clamp maps $\mathbb{R}$ continuously and monotonically into I. The inverse of an order isomorphism between compact real intervals is itself an order isomorphism, hence is continuous and strictly increasing. Their composition is therefore continuous and monotone on the whole line and takes values in I. If $y \in I$, then $\mathrm{clamp}_{[0,1]}(y) = y$, so the usual right-inverse identity for F_I^{-1} gives $F(G(y)) = y$. Likewise, $F(x) \in I$ for $x \in I$, and the left-inverse identity gives $G(F(x)) = x$. Strict increase and bijectivity on I are inherited from F_I^{-1}. □

The totalization is not cosmetic. It produces a globally continuous monotone function and makes the reflection law valid without restricting the variable to $[0,1]$.

2.2 A four-way comparison calculus

The order-isomorphism viewpoint turns inverse inequalities into direct inequalities without any numerical inversion.

Theorem 2.3 (Comparison equivalences; Lean)**.** *For $x, y \in [0,1]$,*

$$G(y) \le x \iff y \le F(x), \tag{3}$$
$$F(x) \le y \iff x \le G(y), \tag{4}$$
$$G(y) < x \iff y < F(x), \tag{5}$$
$$F(x) < y \iff x < G(y). \tag{6}$$

Proof. Apply the strictly increasing function F to the two sides of an inequality involving $G(y)$, then use $F(G(y)) = y$. For example,

$$G(y) \le x \iff F(G(y)) \le F(x) \iff y \le F(x).$$

The other three equivalences are identical, with the order reversed in placement but not in direction because F and G are increasing. □

Remark 2.4. The equivalences are often more useful than an explicit inverse formula. They let one convert a known dyadic or asymptotic bound for F into a location bound for G with no loss in constants.

2.3 Tails, endpoints, and reflection

Theorem 2.5 (Constant tails and symmetry; Lean)**.** *For every $y \in \mathbb{R}$,*

$$y \le 0 \Longrightarrow G(y) = 0, \tag{7}$$
$$y \ge 1 \Longrightarrow G(y) = 1, \tag{8}$$
$$G(1-y) = 1 - G(y). \tag{9}$$

In particular,

$$G(0) = 0, \qquad G(1) = 1, \qquad G(1/2) = 1/2.$$

Proof. The two tail identities follow immediately from the clamp. Suppose first that $0 \le y \le 1$. Then $G(y) \in [0,1]$, and by the symmetry of F,

$$F(1 - G(y)) = 1 - F(G(y)) = 1 - y.$$

The point $1 - G(y)$ lies in $[0,1]$, so uniqueness of the inverse gives $G(1-y) = 1 - G(y)$. If $y \le 0$ or $y \ge 1$, the same equation follows from the two constant tails. Setting $y = 0, 1$, and $1/2$ gives the stated special values. □

3 Exact inversion of the dyadic evaluator

3.1 The exact statement

Let $\mathsf{F}_n(a)$ denote the exact bounded dyadic evaluator from the formal development: for $0 \le a \le 2^n$,

$$\mathsf{F}_n(a) = F\left(\frac{a}{2^n}\right),$$

and for larger a it is clamped at the endpoint value 1.

Theorem 3.1 (Dyadic inversion; Lean). *For all $n, a \in \mathbb{N}$,*

$$G(\mathsf{F}_n(a)) = \frac{\min\{a, 2^n\}}{2^n}.$$

In particular, if $0 \le a \le 2^n$, then

$$G\left(F\left(\frac{a}{2^n}\right)\right) = \frac{a}{2^n}.$$

Proof. When $a \le 2^n$, the argument $a/2^n$ lies in I, so the left-inverse identity in theorem 2.2 applies. When $a \ge 2^n$, the bounded evaluator equals 1, the minimum equals 2^n, and both sides are 1. □

The theorem is stronger than a table of examples: every exact rational produced by the dyadic algorithm immediately gives an exact algebraic input for the inverse.

3.2 A small exact table

x	$F(x)$	exact inverse statement
0	0	$G(0) = 0$
$\frac{1}{16}$	$\frac{143}{2073600}$	$G\left(\frac{143}{2073600}\right) = \frac{1}{16}$
$\frac{1}{8}$	$\frac{1}{288}$	$G\left(\frac{1}{288}\right) = \frac{1}{8}$
$\frac{1}{4}$	$\frac{5}{72}$	$G\left(\frac{5}{72}\right) = \frac{1}{4}$
$\frac{3}{8}$	$\frac{73}{288}$	$G\left(\frac{73}{288}\right) = \frac{3}{8}$
$\frac{1}{2}$	$\frac{1}{2}$	$G\left(\frac{1}{2}\right) = \frac{1}{2}$
$\frac{5}{8}$	$\frac{215}{288}$	$G\left(\frac{215}{288}\right) = \frac{5}{8}$
$\frac{3}{4}$	$\frac{67}{72}$	$G\left(\frac{67}{72}\right) = \frac{3}{4}$
$\frac{7}{8}$	$\frac{287}{288}$	$G\left(\frac{287}{288}\right) = \frac{7}{8}$
$\frac{15}{16}$	$\frac{2073457}{2073600}$	$G\left(\frac{2073457}{2073600}\right) = \frac{15}{16}$

Table 2: Exact inverse values obtained from the dyadic evaluator and reflection.

Remark 3.2. The threshold $F(1/4) = 5/72$ will reappear in the effective endpoint steepness theorem. Its occurrence is structural rather than arbitrary: $1/4 = 2^{-2}$ is precisely the scale on which the quadratic flatness bound is valid.

4 Flatness of F becomes steepness of G

4.1 The transported power inequalities

Write

$$C_n = 2^{\binom{n+1}{2}}.$$

The effective flatness theorem for F says, in one of its standard forms, that

$$0 \le x \le 2^{-n} \quad \Longrightarrow \quad F(x) \le C_n x^n.$$

The sharp version improves the right-hand side by the factor $n!$.

Theorem 4.1 (Inverted effective and sharp flatness; Lean)**.** *Let $n \in \mathbb{N}$ and $0 \le y \le 1$. If*

$$2^n G(y) \le 1,$$

then

$$y \le C_n G(y)^n, \tag{10}$$

$$y \le \frac{C_n}{n!} G(y)^n. \tag{11}$$

Moreover, the scale condition follows from the argument condition

$$0 \le y \le F(2^{-n}).$$

Proof. Put $x = G(y)$. The scale hypothesis says $0 \le x \le 2^{-n}$, while $F(x) = y$. Substituting x into the effective and sharp flatness bounds for F gives (10) and (11).

Now suppose $y \le F(2^{-n})$. By (3), $G(y) \le 2^{-n}$, which is exactly the required scale condition. □

For $n \ge 1$, taking positive nth roots yields the more geometric form.

Corollary 4.2 (Root lower bounds; Lean input, Derived form)**.** *If $n \ge 1$, $0 < y \le F(2^{-n})$, then*

$$G(y) \ge C_n^{-1/n} y^{1/n}, \tag{12}$$

$$G(y) \ge \left(\frac{n!}{C_n}\right)^{1/n} y^{1/n}. \tag{13}$$

Proof. All quantities are nonnegative. Rearrange equations (10) and (11) and apply the monotonicity of the positive nth-root function. □

4.2 An effective bound against every line

The quadratic case already detects infinite endpoint slope.

Theorem 4.3 (Effective linear steepness; Lean)**.** *Let $M, y \in \mathbb{R}$. If*

$$0 < y, \qquad y \le F(1/4) = \frac{5}{72}, \qquad 8M^2 y < 1,$$

then

$$My < G(y).$$

Proof. The comparison theorem gives $G(y) \le 1/4$, so the quadratic inverse flatness estimate applies:

$$y \le 8G(y)^2.$$

If $M \le 0$, then $My \le 0 < G(y)$ because $y > 0$ and G is strictly increasing on $[0, 1]$. Suppose $M > 0$ and, toward a contradiction, $G(y) \le My$. Then

$$y \le 8G(y)^2 \le 8M^2 y^2.$$

Dividing by $y > 0$ gives $1 \le 8M^2 y$, contrary to the last hypothesis. □

4.3 Infinite one-sided slope

Theorem 4.4 (Endpoint quotient divergence; Lean). *As $y \downarrow 0$,*

$$\frac{G(y)}{y} \longrightarrow +\infty.$$

By reflection, as $y \uparrow 1$,

$$\frac{1 - G(y)}{1 - y} \longrightarrow +\infty.$$

Proof. Fix a real target A. Put $B = |A| + 1$. For all sufficiently small positive y we have simultaneously

$$y < F(1/4), \qquad 8B^2 y < 1.$$

By theorem 4.3, $By < G(y)$, and hence

$$A \le B < \frac{G(y)}{y}.$$

This is exactly the filter definition of divergence to $+\infty$.

For the right endpoint use (9):

$$\frac{1 - G(y)}{1 - y} = \frac{G(1 - y)}{1 - y},$$

and let $1 - y \downarrow 0$. □

4.4 Failure of every positive Hölder estimate

Theorem 4.5 (Stronger than non-Lipschitz; Derived). *For every $\alpha > 0$,*

$$\frac{G(y)}{y^\alpha} \longrightarrow +\infty \qquad (y \downarrow 0).$$

Consequently there are no $C, \delta > 0$ for which

$$G(y) \le C y^\alpha \qquad (0 \le y \le \delta).$$

The reflected assertion holds at 1:

$$\frac{1 - G(y)}{(1 - y)^\alpha} \longrightarrow +\infty \qquad (y \uparrow 1).$$

Proof. Choose an integer $n > 1/\alpha$. On the window $0 < y \le F(2^{-n})$, equation (12) gives

$$\frac{G(y)}{y^\alpha} \ge C_n^{-1/n} y^{1/n - \alpha}.$$

The exponent $1/n - \alpha$ is negative, so the right-hand side tends to $+\infty$. A local Hölder upper bound would keep the quotient bounded and is therefore impossible. The statement at 1 follows from reflection. □

Each fixed n supplies a root lower bound only on the shrinking window $y \le F(2^{-n})$. There is no bound uniform in n. This is not a defect: it is exactly the scale dependence needed to prove that G outruns *every fixed* positive power.

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5 Classical calculus and the exact regularity of the inverse

This section uses the formalized inverse package together with standard one-variable analysis. Smoothness, the reciprocal first-derivative formula, and positivity are now marked LEAN; the second and higher derivative formulas remain DERIVED.

5.1 Smoothness and differential identities in the interior

Theorem 5.1 (Interior inverse calculus through first order; LEAN)**.** *The inverse G is C^∞ on $(0,1)$. For $0<y<1$,*

$$G'(y)=\frac{1}{F'(G(y))}=\frac{1}{2\,\mathrm{up}(2G(y)-1)}>0. \tag{14}$$

Proof. For $x\in(0,1)$, (2) and positivity of up on $(-1,1)$ imply $F'(x)>0$. The C^∞ inverse function theorem therefore applies at every interior point and gives a smooth local inverse. Uniqueness identifies all these local inverses with G. Differentiating $F(G(y))=y$ and substituting (2) proves (14). □

The corresponding Lean declarations are `fabiusInv_contDiffOn_Ioo`, `fabiusInv_hasDerivAt`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`.

Proposition 5.2 (Higher inverse derivatives; DERIVED)**.** *For $0<y<1$,*

$$G''(y)=-\frac{F''(G(y))}{F'(G(y))^3}. \tag{15}$$

More generally, every derivative of G is a rational polynomial in the derivatives of F evaluated at $G(y)$, with a power of $F'(G(y))$ in the denominator.

Proof. Differentiating the identity behind (14) once more gives

$$F''(G(y))G'(y)^2+F'(G(y))G''(y)=0.$$

Substitution of (14) yields (15). Repeated differentiation gives the higher formulas, equivalently the Faà di Bruno inverse-derivative polynomials. □

Corollary 5.3 (Midpoint differential data; DERIVED)**.** *At the fixed point $1/2$,*

$$G(1/2)=1/2,\qquad G'(1/2)=\frac{1}{2},\qquad G''(1/2)=0.$$

Proof. The value follows from reflection. Since $\mathrm{up}(0)=1$, (14) gives $G'(1/2)=1/2$. The up-function is even, so $\mathrm{up}'(0)=0$, hence $F''(1/2)=4\,\mathrm{up}'(0)=0$; now use (15). □

5.2 Shape reversal under inversion

The formal development proves that F is strictly convex on $(0,1/2)$ and strictly concave on $(1/2,1)$. Inversion reverses these two curvature signs.

Proposition 5.4 (Concavity and convexity of the inverse; DERIVED)**.** *The function G is strictly concave on $(0,1/2)$ and strictly convex on $(1/2,1)$. Its tangent slope decreases from $+\infty$ to $1/2$ on the left half, then increases from $1/2$ to $+\infty$ on the right half.*

Proof. For $0<y<1/2$, strict increase gives $0<G(y)<1/2$, where $F''>0$. Equation (15) and $F'>0$ imply $G''(y)<0$. Similarly, $y>1/2$ gives $G(y)>1/2$, where $F''<0$, and therefore $G''(y)>0$.

The endpoint divergence of the derivative is understood as a one-sided slope statement: by theorem 4.4, the secant slopes from the origin diverge, and concavity makes the right derivative at the origin at least as large as every such limiting secant slope. Reflection gives the right endpoint. The midpoint slope is corollary 5.3. □

5.3 The C^∞ and analytic loci

Theorem 5.5 (Exact regularity locus; DERIVED)**.** *The totalized inverse has the following regularity.*

1. *G is C^∞ precisely on*
$$\mathbb{R} \setminus \{0, 1\}.$$
2. *It has no finite derivative at 0 or at 1.*
3. *It is real analytic precisely on*
$$\mathbb{R} \setminus [0, 1].$$

Proof. On $(-\infty, 0)$ and $(1, \infty)$ the function is constant, hence smooth and analytic. On $(0, 1)$ smoothness follows from theorem 5.1. At 0, the left derivative is 0 because the function is constant to the left, whereas the right difference quotient $G(y)/y$ tends to $+\infty$. Thus no finite derivative exists. Reflection gives the same conclusion at 1.

It remains to exclude analyticity in the open unit interval. Suppose G were analytic near some $y_0 \in (0, 1)$. Its derivative there is nonzero by (14); the analytic inverse function theorem would then make its local inverse F analytic near $x_0 = G(y_0)$. But the Fabius function is nowhere analytic on the interior of its transition interval. This contradiction proves the claim. □

6 Inverting the sharp small-argument asymptotic

The power inequalities show qualitative infinite steepness. The sharp Lambert expansion gives the actual scale. This section extracts that scale without needing the detailed value of the periodic fluctuation.

6.1 The exact lower Lambert phase

Put
$$L = \log 2.$$

For sufficiently small $x > 0$, let $\lambda = \lambda(x)$ be the large positive solution of
$$x = \lambda 2^{-\lambda} = \lambda \mathrm{e}^{-L\lambda}. \tag{16}$$

Equivalently,
$$\lambda(x) = -\frac{1}{L} W_{-1}(-Lx),$$
but (16) is the identity used in the proofs.

The sharp expansion established for F has the form
$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) + \mathcal{O}(\lambda^{-1}), \tag{17}$$

where Ψ is smooth, bounded, and one-periodic. The exact value of C_F and the Fourier description of Ψ are important for F itself, but surprisingly they do not affect the leading equivalent of the inverse.

6.2 Asymptotic solution for the phase

Let $y = F(x)$ and
$$T = \log \frac{1}{y}.$$
Negating (17) gives
$$T = \frac{L}{2}\lambda^2 - B\lambda + \frac{1}{2}\log\lambda - C_F - \Psi(\lambda) + \mathcal{O}(\lambda^{-1}), \qquad B = 1 + \frac{L}{2}. \tag{18}$$

Lemma 6.1 (Asymptotic inversion of the quadratic phase; Derived)**.** *As $T \to +\infty$,*
$$\lambda = \sqrt{\frac{2T}{L}} + \frac{B}{L} + \mathcal{O}\left(\frac{\log T}{\sqrt{T}}\right).$$

Proof. Since Ψ is bounded, (18) first gives
$$T = \frac{L}{2}\lambda^2 + \mathcal{O}(\lambda + \log\lambda).$$
Hence $\lambda \to \infty$ and, with
$$q = \sqrt{\frac{2T}{L}},$$
we have $\lambda/q \to 1$. Dividing
$$T - \frac{L}{2}\lambda^2 = -\frac{L}{2}(\lambda - q)(\lambda + q)$$
by $\lambda + q \asymp q$ and using the preceding error estimate improves this to $\lambda = q + \mathcal{O}(1)$.

Write
$$\lambda = q + \frac{B}{L} + \rho.$$
Insert this into (18). The terms proportional to q cancel because $L(B/L)q = Bq$. What remains is
$$Lq\rho = -\frac{1}{2}\log\left(q + \frac{B}{L} + \rho\right) + \mathcal{O}(1) + \mathcal{O}(\rho^2).$$
The already known bound $\rho = \mathcal{O}(1)$ makes the final term bounded. Since $q \asymp \sqrt{T}$ and $\log q \asymp \log T$, division by Lq gives
$$\rho = \mathcal{O}\left(\frac{\log T}{\sqrt{T}}\right).$$
□

6.3 The endpoint equivalent for the inverse

Theorem 6.2 (Sharp inverse asymptotic; Derived)**.** *As $y \downarrow 0$,*
$$\boxed{G(y) \sim \frac{\sqrt{\log(1/y)}}{\mathrm{e}\sqrt{\log 2}} \exp\left[-\sqrt{2\log 2\,\log(1/y)}\right].} \tag{19}$$

Equivalently,
$$\log G(y) = -\sqrt{2L\log(1/y)} + \frac{1}{2}\log\log(1/y) - 1 - \frac{1}{2}\log L + o(1). \tag{20}$$

At the right endpoint,
$$1 - G(y) \sim \frac{\sqrt{\log(1/(1-y))}}{\mathrm{e}\sqrt{\log 2}} \exp\left[-\sqrt{2\log 2\,\log(1/(1-y))}\right] \quad (y \uparrow 1). \tag{21}$$

Proof. Let $x = G(y)$. Then $y = F(x)$ and x is related to its exact phase by $x = \lambda e^{-L\lambda}$. Therefore

$$\log x = \log\lambda - L\lambda.$$

With $T = \log(1/y)$, lemma 6.1 gives

$$\lambda = q + \frac{B}{L} + o(1), \qquad q = \sqrt{\frac{2T}{L}},$$

because $(\log T)/\sqrt{T} = o(1)$. Also $\log\lambda = \log q + o(1)$. Consequently

$$\begin{aligned}\log x &= \log q - Lq - B + o(1)\\ &= -\sqrt{2LT} + \frac{1}{2}\log T + \frac{1}{2}\log\frac{2}{L} - 1 - \frac{L}{2} + o(1)\\ &= -\sqrt{2LT} + \frac{1}{2}\log T - 1 - \frac{1}{2}\log L + o(1),\end{aligned}$$

because $L = \log 2$. This is (20). Exponentiating proves (19). Finally, (9) gives $1 - G(y) = G(1-y)$ and hence (21). □

Remark 6.3 (Why the periodic fluctuation disappears at leading order)*.* The bounded term $\Psi(\lambda)$ changes the solution λ of (18) only by $\mathcal{O}(1/\lambda)$. Since $\log x = \log\lambda - L\lambda$, this changes $\log x$ by $o(1)$ and therefore changes x by a factor tending to 1. Periodicity returns in the next relative correction, but not in the leading equivalent (19).

6.4 The complete hierarchy of elementary scales

Corollary 6.4 (Between powers and logarithms; Derived)**.** *For every $\alpha > 0$ and every $m > 0$,*

$$y^\alpha = o(G(y)), \tag{22}$$
$$G(y) = o\big((\log(1/y))^{-m}\big) \tag{23}$$

as $y \downarrow 0$.

Proof. Set $T = \log(1/y)$. Formula (20) gives

$$\log\frac{G(y)}{y^\alpha} = \alpha T - \sqrt{2LT} + \mathcal{O}(\log T) \longrightarrow +\infty,$$

which proves (22). Likewise,

$$\log(G(y)T^m) = -\sqrt{2LT} + \mathcal{O}(\log T) \longrightarrow -\infty,$$

which proves (23). □

The inverse endpoint is therefore neither power-like nor logarithmic. It lies in the stretched-logarithmic regime $\exp(-c\sqrt{\log(1/y)})$ with an essential $\sqrt{\log(1/y)}$ prefactor.

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7 Closed periodic saddle jets

We now turn from the inverse to the coefficient machinery hidden inside the all-orders endpoint expansion. Throughout this part,

$$L = \log 2, \qquad D = \frac{\mathrm{d}}{\mathrm{d}u},$$

and Ψ is the smooth one-periodic function occurring in the sharp Lambert main term.

7.1 The recursive jets

Define the periodic jets p_n by

$$p_0(u) = \frac{1}{2} + \frac{\Psi'(u)}{L}, \tag{24}$$

$$p_{n+1}(u) = \left(\frac{D}{L} - (n+1)\right) p_n(u) + \frac{(-1)^{n+1} n!}{L}. \tag{25}$$

The bounded exponent jets are

$$d_n(u) = p_n(u) + (-1)^{n+1} n!. \tag{26}$$

The subtraction in (26) removes the factorial part that would otherwise obscure the centered saddle expansion.

For $n \geq 0$, write

$$H_n = \sum_{k=1}^{n} \frac{1}{k}, \qquad H_0 = 0,$$

and define integers $c(n,m)$ by

$$\prod_{k=1}^{n} (z-k) = \sum_{m=0}^{n} c(n,m) z^m. \tag{27}$$

Theorem 7.1 (Closed periodic jets; Lean)**.** *For every $n \geq 0$,*

$$p_n(u) = (-1)^n n! \left(\frac{1}{2} + \frac{H_n}{L}\right) + \sum_{m=0}^{n} c(n,m) \frac{\Psi^{(m+1)}(u)}{L^{m+1}}, \tag{28}$$

$$d_n(u) = (-1)^n n! \left(\frac{H_n}{L} - \frac{1}{2}\right) + \sum_{m=0}^{n} c(n,m) \frac{\Psi^{(m+1)}(u)}{L^{m+1}}. \tag{29}$$

Proof. Let

$$T_k = \frac{D}{L} - k.$$

The operators T_k commute because each is a polynomial in D. Iterating (25) gives

$$p_n = T_n T_{n-1} \cdots T_1 p_0 + \sum_{j=0}^{n-1} T_n T_{n-1} \cdots T_{j+2} \left(\frac{(-1)^{j+1} j!}{L}\right), \tag{30}$$

where the product is empty when $j = n-1$.

First consider the homogeneous term. Since

$$T_n \cdots T_1 = \prod_{k=1}^{n} \left(\frac{D}{L} - k\right),$$

its action on the constant $1/2$ is

$$\frac{1}{2} \prod_{k=1}^{n} (-k) = \frac{(-1)^n n!}{2}.$$

Its action on Ψ'/L is found by expanding the operator polynomial with (27):

$$\prod_{k=1}^{n} \left(\frac{D}{L} - k\right) \frac{\Psi'}{L} = \sum_{m=0}^{n} c(n,m) \frac{\Psi^{(m+1)}}{L^{m+1}}.$$

Now propagate the forcing term inserted at stage j. It is constant, so each subsequent T_k acts on it as multiplication by $-k$. Hence its contribution at stage n is

$$\frac{(-1)^{j+1}j!}{L}\prod_{k=j+2}^{n}(-k) = \frac{(-1)^{j+1}j!}{L}\frac{(-1)^{n-j-1}n!}{(j+1)!}$$
$$= \frac{(-1)^n n!}{(j+1)L}.$$

Summing over $j = 0, \dots, n-1$ gives

$$\frac{(-1)^n n!}{L}\sum_{j=0}^{n-1}\frac{1}{j+1} = \frac{(-1)^n n! H_n}{L}.$$

Combining the constant, derivative, and forcing contributions proves (28). Finally,

$$p_n + (-1)^{n+1}n! = (-1)^n n!\left(\frac{1}{2} + \frac{H_n}{L} - 1\right) + \text{derivative part},$$

which is exactly (29). □

7.2 Why the Stirling indices are shifted

Let $s(n,k)$ be the signed Stirling number of the first kind, defined by

$$z(z-1)\cdots(z-n+1) = \sum_{k=0}^{n} s(n,k)z^k.$$

Proposition 7.2 (Shifted Stirling conversion; Lean)**.** *For* $0 \le m \le n$*,*

$$c(n,m) = s(n+1, m+1).$$

Proof. Apply the defining identity with $n+1$:

$$z(z-1)\cdots(z-n) = \sum_{k=0}^{n+1} s(n+1,k)z^k.$$

Both sides have zero constant term, so divide by z:

$$(z-1)\cdots(z-n) = \sum_{m=0}^{n} s(n+1,m+1)z^m.$$

Comparison with (27) proves the formula. □

Warning 7.3*.* The coefficients are not $s(n,m)$. The product starts with $(z-1)$ rather than z, and this forces the simultaneous shift $(n,m) \mapsto (n+1,m+1)$. In particular, the nonzero constant column $c(n,0) = (-1)^n n!$ is essential.

7.3 Coefficient rows and the first jets

The first rows of $c(n,m)$, listed from $m = 0$ to $m = n$, are

n	$c(n,0)$	$c(n,1)$	$c(n,2)$	$c(n,3)$	$c(n,4)$	$c(n,5)$
0	1					
1	−1	1				
2	2	−3	1			
3	−6	11	−6	1		
4	24	−50	35	−10	1	
5	−120	274	−225	85	−15	1

and (29) gives

$$d_0 = -\frac{1}{2} + \frac{\Psi'}{L}, \tag{31}$$

$$d_1 = \frac{1}{2} - \frac{1}{L} - \frac{\Psi'}{L} + \frac{\Psi''}{L^2}, \tag{32}$$

$$d_2 = -1 + \frac{3}{L} + \frac{2\Psi'}{L} - \frac{3\Psi''}{L^2} + \frac{\Psi'''}{L^3}, \tag{33}$$

$$d_3 = 3 - \frac{11}{L} - \frac{6\Psi'}{L} + \frac{11\Psi''}{L^2} - \frac{6\Psi'''}{L^3} + \frac{\Psi^{(4)}}{L^4}. \tag{34}$$

Corollary 7.4 (Structural properties; Lean)**.** *Every p_n and d_n is smooth, one-periodic, and bounded on $\mathbb{R}$. Moreover, d_n involves derivatives of Ψ only through order $n+1$, and the coefficient of the highest derivative is $L^{-(n+1)}$.*

Proof. The closed formulas are finite linear combinations of derivatives of the smooth periodic function Ψ and constants. Smoothness, periodicity, and boundedness follow. Since $c(n,n) = 1$, the coefficient of $\Psi^{(n+1)}$ is $L^{-(n+1)}$. □

8 The centered exponent and its algebraic collapse

8.1 From a long Taylor expansion to one short formula

At the saddle, introduce the half-power parameter

$$\varepsilon = \lambda^{-1/2}$$

and the Gaussian variable v. After removing the exact quadratic term, the centered exponent has a formal expansion

$$\mathcal{E}(u, v; \varepsilon) = \sum_{m \geq 1} E_m(u, v)\varepsilon^m.$$

The original construction of E_m contains separate contributions from the vertical negative-Laplace jet and from the Taylor expansion of $\log(1 + \mathrm{i}\varepsilon v)$. The bounded exponent jets make these two pieces collapse.

Theorem 8.1 (Closed exponent coefficient; Lean)**.** *For every $m \geq 1$,*

$$E_m(u, v) = \mathrm{i}^m \left(\frac{d_{m-1}(u)}{m!} v^m + \frac{(-1)^{m+1}}{m+2} v^{m+2} \right). \tag{35}$$

In particular,

$$E_1 = \mathrm{i}v \, \frac{3d_0 + v^2}{3}, \tag{36}$$

$$E_2 = v^2 \, \frac{-2d_1 + v^2}{4}, \tag{37}$$

$$E_3 = \mathrm{i}v^3 \left(-\frac{d_2}{6} - \frac{v^2}{5} \right), \tag{38}$$

$$E_4 = \frac{v^4}{24}(d_3 - 4v^2). \tag{39}$$

Proof. Before centering, the coefficient at order m has two contributions. The vertical jet contributes $\mathrm{i}^m p_{m-1}(u)v^m/m!$, while the logarithmic saddle coordinate contributes a factorial term to the same monomial and a universal v^{m+2} term. Using

$$d_{m-1} = p_{m-1} + (-1)^m(m-1)!$$

combines the two v^m coefficients into $\mathrm{i}^m d_{m-1}/m!$. The remaining logarithmic term is

$$\frac{\mathrm{i}^{m+2}(-1)^m}{m+2}v^{m+2} = \mathrm{i}^m\frac{(-1)^{m+1}}{m+2}v^{m+2},$$

because $\mathrm{i}^2 = -1$. This is (35). Substitution of $m = 1, 2, 3, 4$ gives (36)–(39). ☐

Corollary 8.2 (Parity; Lean)**.** *For every $m \geq 1$,*

$$E_m(u, -v) = (-1)^m E_m(u, v).$$

Thus the odd-indexed E_m are odd polynomials in v and the even-indexed E_m are even.

Proof. Both monomials in (35) have parity m, because $m+2$ has the same parity as m. ☐

9 Formal exponential, Gaussian contraction, and logarithm

9.1 The exponential recursion

Define polynomials $B_n(u, v)$ by the formal identity

$$\exp\left(\sum_{m\geq 1} E_m(u, v)\varepsilon^m\right) = \sum_{n\geq 0} B_n(u, v)\varepsilon^n. \tag{40}$$

Thus $B_0 = 1$.

Proposition 9.1 (Finite exponential recursion; Lean)**.** *For $n \geq 1$,*

$$nB_n = \sum_{m=1}^{n} mE_m B_{n-m}. \tag{41}$$

Equivalently,

$$B_n = \sum_{\substack{r\geq 1\\ m_1+\cdots+m_r=n}} \frac{1}{r!}E_{m_1}\cdots E_{m_r},$$

where the inner sum is over ordered compositions of n into r positive parts.

Proof. Differentiate (40) with respect to ε. The derivative of the left side is

$$\left(\sum_{m\geq 1} mE_m\varepsilon^{m-1}\right)\left(\sum_{k\geq 0} B_k\varepsilon^k\right),$$

whereas the derivative of the right side is $\sum_{n\geq 1} nB_n\varepsilon^{n-1}$. Comparing the coefficient of ε^{n-1} gives (41). Expanding the exponential as

$$\sum_{r\geq 0}\frac{1}{r!}\left(\sum_{m\geq 1} E_m\varepsilon^m\right)^r$$

and collecting total degree n gives the composition formula. ☐

Corollary 9.2 (Parity of the exponential coefficients; Lean)**.** *For all $n \geq 0$,*

$$B_n(u, -v) = (-1)^n B_n(u, v).$$

Proof. Every product contributing to B_n has indices $m_1 + \cdots + m_r = n$. By corollary 8.2, its parity is $(-1)^{m_1+\cdots+m_r} = (-1)^n$. ☐

9.2 Gaussian contraction and mass coefficients

For a polynomial P define normalized Gaussian contraction by

$$\mathfrak{G}[P] = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \mathrm{e}^{-v^2/2} P(v)\, \mathrm{d}v.$$

It is determined by

$$\mathfrak{G}[v^{2k}] = (2k-1)!!, \qquad \mathfrak{G}[v^{2k+1}] = 0. \tag{42}$$

The mass coefficients are

$$a_j(u) = \mathfrak{G}[B_{2j}(u,\cdot)], \qquad j \geq 0. \tag{43}$$

In particular $a_0 = 1$. The odd half-powers vanish automatically:

$$\mathfrak{G}[B_{2j+1}] = 0$$

by corollary 9.2.

Theorem 9.3 (Closed composition formula for a_j; Derived)**.** *For $j \geq 1$,*

$$a_j(u) = (-1)^j \sum_{r=1}^{2j} \frac{1}{r!} \sum_{\substack{m_1+\cdots+m_r=2j \\ m_i \geq 1}} \sum_{S \subseteq \{1,\ldots,r\}} (2j+2|S|-1)!! \\ \times \prod_{i \in S} \frac{(-1)^{m_i+1}}{m_i+2} \prod_{i \notin S} \frac{d_{m_i-1}(u)}{m_i!}. \tag{44}$$

Consequently a_j is a polynomial with rational coefficients in $d_0, \ldots, d_{2j-1}$.

Proof. Use the ordered-composition formula in proposition 9.1 for B_{2j}. Each factor

$$E_{m_i} = \mathrm{i}^{m_i} \left(\frac{d_{m_i-1}}{m_i!} v^{m_i} + \frac{(-1)^{m_i+1}}{m_i+2} v^{m_i+2} \right)$$

offers two choices. Let S be the set of factors from which the second, universal monomial is chosen. Since $\sum_i m_i = 2j$, the product of the powers of i is

$$\mathrm{i}^{2j} = (-1)^j,$$

and the total power of v is $2j + 2|S|$. Its Gaussian contraction is $(2j+2|S|-1)!!$. Multiplying the remaining scalar factors gives (44). The largest possible jet index is $m_i - 1 \leq 2j - 1$. $\square$

Formula (44) is a human-readable algebraic consequence of the formal exponent and exponential-recursion theorems. The repository machine-checks the underlying recursion, the structural top-jet theorem, and explicit low orders. It does not package this exact threefold summation as a single general Lean declaration.

9.3 Logarithmic coefficients

Define $A_j(u)$ by

$$1+\sum_{j\geq 1} a_j(u)z^j = \exp\left(\sum_{j\geq 1} A_j(u)z^j\right). \tag{45}$$

These are the coefficients that occur in $\log F$.

Proposition 9.4 (Mass-to-logarithm conversion; Lean)**.** *For $n \geq 1$,*

$$A_n = a_n - \frac{1}{n}\sum_{k=1}^{n-1} kA_k\, a_{n-k}. \tag{46}$$

Equivalently,

$$A_n = \sum_{r=1}^{n} \frac{(-1)^{r-1}}{r} \sum_{\substack{q_1+\cdots+q_r=n \\ q_i\geq 1}} a_{q_1}\cdots a_{q_r}. \tag{47}$$

Proof. Differentiate (45) logarithmically:

$$\sum_{n\geq 1} na_n z^{n-1} = \left(1+\sum_{j\geq 1} a_j z^j\right)\left(\sum_{k\geq 1} kA_k z^{k-1}\right).$$

The coefficient of z^{n-1} gives

$$na_n = \sum_{k=1}^{n} kA_k a_{n-k}, \qquad a_0 = 1,$$

and solving for A_n proves (46). Formula (47) is the coefficient of z^n in the ordinary series

$$\log(1+w) = \sum_{r\geq 1} \frac{(-1)^{r-1}}{r} w^r, \qquad w = \sum_{j\geq 1} a_j z^j.$$

□

Corollary 9.5 (Regularity and derivative order; Lean input, Derived conclusion)**.** *For every j, the functions a_j and A_j are smooth, one-periodic, and bounded. Neither involves a derivative of Ψ of order greater than $2j$.*

Proof. The function a_j is a polynomial in $d_0, \ldots, d_{2j-1}$, and d_n is smooth, periodic, and depends on Ψ only through order $n+1$. Thus a_j has the stated properties and uses derivatives only through order $2j$. The recursion (46) transfers the same assertions to A_j. □

9.4 The top jet and the highest derivative

The most complicated-looking coefficient has a very simple dependence on its newest jet.

Theorem 9.6 (Leading jet law; Lean)**.** *For $j \geq 1$, the mass coefficient a_j is affine in d_{2j-1}, and the coefficient of that jet is*

$$\frac{(-1)^j}{2^j j!}. \tag{48}$$

The logarithmic coefficient A_j has the same top-jet coefficient. Consequently the coefficient of $\Psi^{(2j)}(u)$ in $A_j(u)$ is

$$\boxed{\frac{(-1)^j}{2^j j! L^{2j}}.} \tag{49}$$

Proof. The jet d_{2j-1} occurs only in E_{2j}. Therefore the only term of B_{2j} that contains it is the linear term E_{2j} itself. From (35), its jet part is

$$\mathrm{i}^{2j}\frac{d_{2j-1}}{(2j)!}v^{2j} = (-1)^j\frac{d_{2j-1}}{(2j)!}v^{2j}.$$

Gaussian contraction multiplies this by $(2j-1)!!$. Since

$$(2j)! = 2^j j!(2j-1)!!,$$

the coefficient is (48).

In (46), every correction term $A_k a_{j-k}$ has $k < j$ and $j - k < j$; it involves jets only through d_{2j-3} and therefore cannot alter the coefficient of d_{2j-1}. Thus A_j has the same top-jet coefficient. Finally, corollary 7.4 says that the coefficient of $\Psi^{(2j)}$ in d_{2j-1} is L^{-2j}, giving (49). $\square$

Remark 9.7. The theorem provides a quick consistency check for every explicit coefficient. For example, A_1 must contain $-\Psi''/(2L^2)$, while A_2 must contain $+\Psi^{(4)}/(8L^4)$.

10 The first and second logarithmic corrections

10.1 Order two: the first correction

From equations (36) and (37),

$$B_2 = E_2 + \frac{1}{2}E_1^2.$$

A direct expansion gives

$$B_2 = -\frac{d_0^2 + d_1}{2}v^2 + \left(\frac{1}{4} - \frac{d_0}{3}\right)v^4 - \frac{1}{18}v^6.$$

Using $\mathfrak{G}[v^2] = 1$, $\mathfrak{G}[v^4] = 3$, and $\mathfrak{G}[v^6] = 15$ yields

$$a_1 = A_1 = -\frac{d_0^2}{2} - d_0 - \frac{d_1}{2} - \frac{1}{12}. \tag{50}$$

Equivalently,

$$A_1 = -\frac{6d_0^2 + 12d_0 + 6d_1 + 1}{12}.$$

Substituting equations (31) and (32) recovers the first periodic correction in terms of Ψ' and Ψ''.

10.2 Order four of the formal exponential

The fourth exponential coefficient is

$$B_4 = E_4 + E_3E_1 + \frac{1}{2}E_2^2 + \frac{1}{2}E_2E_1^2 + \frac{1}{24}E_1^4. \tag{51}$$

This follows either from the composition formula or from the recurrence (41). Substitution of equations (36) to (39) and collection of even powers of v gives the following polynomial.

Proposition 10.1 (The order-four reference polynomial; Lean)**.** *Writing* $d_k = d_k(u)$,

$$\begin{aligned}B_4(u,v) = &\left(\frac{d_0^4}{24} + \frac{d_0^2 d_1}{4} + \frac{d_0 d_2}{6} + \frac{d_1^2}{8} + \frac{d_3}{24}\right) v^4 \\ &+ \left(\frac{d_0^3}{18} - \frac{d_0^2}{8} + \frac{d_0 d_1}{6} + \frac{d_0}{5} - \frac{d_1}{8} + \frac{d_2}{18} - \frac{1}{6}\right) v^6 \\ &+ \left(\frac{d_0^2}{36} - \frac{d_0}{12} + \frac{d_1}{36} + \frac{47}{480}\right) v^8 \\ &+ \left(\frac{d_0}{162} - \frac{1}{72}\right) v^{10} + \frac{1}{1944} v^{12}. \end{aligned} \tag{52}$$

Proof. Let

$$P_1 = d_0 v + \frac{v^3}{3}, \qquad P_3 = -\frac{d_2 v^3}{6} - \frac{v^5}{5},$$

so that $E_1 = \mathrm{i}P_1$ and $E_3 = \mathrm{i}P_3$. Then $E_3E_1 = -P_3P_1$ and $E_1^4 = P_1^4$. The even terms E_2, E_4 are already real. Insert these expressions in (51). Each summand has degree at most 12 and is even. Multiplication gives

	v^4	v^6	v^8	v^{10}	v^{12}
E_4	$d_3/24$	$-1/6$	0	0	0
E_3E_1	$d_0d_2/6$	$d_2/18 + d_0/5$	$1/15$	0	0
$E_2^2/2$	$d_1^2/8$	$-d_1/8$	$1/32$	0	0
$E_2E_1^2/2$	$d_0^2d_1/4$	$d_0d_1/6 - d_0^2/8$	$d_1/36 - d_0/12$	$-1/72$	0
$E_1^4/24$	$d_0^4/24$	$d_0^3/18$	$d_0^2/36$	$d_0/162$	$1/1944$

and the two universal v^8 contributions satisfy $1/15 + 1/32 = 47/480$. Summing the columns proves (52). □

10.3 Gaussian contraction and the second mass coefficient

The relevant Gaussian moments are

$$\mathfrak{G}[v^4] = 3, \quad \mathfrak{G}[v^6] = 15, \quad \mathfrak{G}[v^8] = 105, \quad \mathfrak{G}[v^{10}] = 945, \quad \mathfrak{G}[v^{12}] = 10395.$$

Theorem 10.2 (Second mass coefficient; Lean)**.** *Gaussian contraction of* (52) *gives*

$$\begin{aligned}a_2 = &\frac{d_0^4}{8} + \frac{5d_0^3}{6} + \frac{3d_0^2 d_1}{4} + \frac{25 d_0^2}{24} + \frac{5 d_0 d_1}{2} + \frac{d_0 d_2}{2} + \frac{d_0}{12} \\ &+ \frac{3d_1^2}{8} + \frac{25 d_1}{24} + \frac{5 d_2}{6} + \frac{d_3}{8} + \frac{1}{288}.\end{aligned} \tag{53}$$

Proof. Multiply the coefficients of $v^4, v^6, v^8, v^{10}, v^{12}$ in (52) by $3, 15, 105, 945, 10395$, respectively, and collect like monomials. For illustration, the coefficient of d_0 is

$$15 \cdot \frac{1}{5} + 105\left(-\frac{1}{12}\right) + 945 \cdot \frac{1}{162} = 3 - \frac{35}{4} + \frac{35}{6} = \frac{1}{12},$$

and the universal constant is

$$15\left(-\frac{1}{6}\right) + 105\frac{47}{480} + 945\left(-\frac{1}{72}\right) + 10395\frac{1}{1944} = \frac{1}{288}.$$

The remaining coefficients follow by the same arithmetic. □

10.4 The second logarithmic coefficient

Since

$$\log(1 + a_1 z + a_2 z^2 + \cdots) = a_1 z + \left(a_2 - \frac{1}{2}a_1^2\right) z^2 + \cdots ,$$

we have $A_2 = a_2 - a_1^2/2$.

Theorem 10.3 (Second periodic saddle correction; Lean)**.** *The second logarithmic coefficient is*

$$\boxed{A_2 = \frac{8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48d_0 d_1 + 12d_0 d_2 + 6d_1^2 + 24d_1 + 20d_2 + 3d_3}{24}.} \tag{54}$$

It is smooth, one-periodic, and bounded.

Proof. Insert (50) and (53) into $A_2 = a_2 - a_1^2/2$. The quartic term $d_0^4/8$ cancels, as do the terms $d_0/12$ and $1/288$ after the square is expanded. The remaining terms reduce to (54). Smoothness, periodicity, and boundedness follow because A_2 is a polynomial in d_0, d_1, d_2, d_3. □

Corollary 10.4 (Two explicit correction orders; Lean)**.** *Let $\lambda = \lambda(x)$ be the exact lower Lambert phase. Then, as $x \downarrow 0$,*

$$\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda)}{\lambda} + \frac{A_2(\lambda)}{\lambda^2} + \mathcal{O}(\lambda^{-3}), \tag{55}$$

where $\mathcal{M}(x)$ is the exact sharp Lambert main term. Since $\lambda \asymp -\log x$, the remainder is also $\mathcal{O}((-\log x)^{-3})$.

Warning 10.5. The coefficients in (55) are evaluated at the exact phase $\lambda(x)$. Substituting a truncated asymptotic formula for λ into A_1 or A_2 is not licensed merely by $\lambda - \lambda_{\text{approx}} = \mathcal{O}(1)$: periodic functions are sensitive to bounded phase shifts.

11 Why the all-orders remainder is actually rigorous

The formal coefficient algebra by itself does not prove an asymptotic expansion. One must control the exact contour integral uniformly on a window that grows with the requested order, prove that exponentiation does not magnify the Taylor error, replace the finite Gaussian window by the whole line, and show that the discarded contour is smaller than every retained inverse power. This section spells out that analytic chain.

11.1 The exact normalized saddle mass

For $x > 0$ small, let $\lambda = \lambda(x)$ be the exact phase (16). The exact saddle reduction can be organized as

$$\log F(x) = \mathcal{M}(x) + \log \mathcal{G}(x) + \mathcal{T}(x), \tag{56}$$

where:

- $\mathcal{M}(x)$ is the exact sharp Lambert main term, containing the quadratic phase, the Gaussian normalization, the periodic finite part, and the fixed constant;
- $\mathcal{G}(x) > 0$ is the normalized saddle-kernel mass and $\mathcal{G}(x) \to 1$;
- $\mathcal{T}(x)$ is the forward negative-Laplace tail, which is flat against every inverse power of λ.

Thus all algebraic corrections beyond $\mathcal{M}$ come from $\log \mathcal{G}$.

It is technically convenient to begin on the real dyadic logarithmic scale

$$x = 2^{-t}, \qquad t \to +\infty,$$

and to write

$$b = \lambda(2^{-t}), \qquad \varepsilon = b^{-1/2}.$$

This is not a restriction to integer t. Every sufficiently small positive x has the unique representation $x = 2^{-t}$ with $t = -\log_2 x$.

After the saddle contour is centered and rescaled by $v = \sqrt{b}\,\theta$, its central kernel has the normalized form

$$\frac{1}{\sqrt{2\pi}} \mathrm{e}^{-v^2/2} \exp R_b(v), \tag{57}$$

where R_b is the exact centered exponent. The formal expansion of R_b is

$$R_b(v) \sim \sum_{m \geq 1} E_m(b, v) b^{-m/2}. \tag{58}$$

The first variable of E_m is evaluated at b because all coefficient functions are periodic in the exact phase.

11.2 The order-dependent central window

For a requested order $N \geq 0$, define

$$A_N(b) = \sqrt{32(N+1)\log b}. \tag{59}$$

Lemma 11.1 (Scale separation; Lean)**.** *For every fixed N,*

$$A_N(b) \longrightarrow +\infty, \qquad \varepsilon A_N(b) \longrightarrow 0 \qquad (b \to +\infty).$$

Moreover, for every fixed $r > 0$ and $\delta > 0$,

$$A_N(b)^r = o(b^\delta).$$

Proof. The first limit is immediate from $\log b \to \infty$. The second follows from

$$\varepsilon A_N(b) = \sqrt{\frac{32(N+1)\log b}{b}} \longrightarrow 0.$$

Finally, every fixed power of $\log b$ grows more slowly than every positive power of b. □

The window has two simultaneous purposes. It grows, so replacing a Gaussian integral over it by a whole-line Gaussian integral creates a tiny tail. Yet its angular width $A_N(b)/\sqrt{b}$ shrinks to zero, so all local Taylor estimates remain uniform.

11.3 All-order control of the exact centered exponent

The exact negative-Laplace factor contains a finite periodic part plus endpoint and forward-product corrections. On the vertical line through the saddle, the proof decomposes it into:

1. a Taylor polynomial in $\theta = v/\sqrt{b}$ whose coefficients are the periodic jets p_n;
2. the Taylor polynomial of $\log(1 + \mathrm{i}\theta)$;
3. two boundary-jet remainders from the finite part;

4. forward-product jets whose size is exponentially small in b;
5. explicit integral remainders for the vertical derivatives.

The factorial pieces in the first two items combine into the bounded jets d_n, producing (35).

Proposition 11.2 (Integrated exponent truncation; Lean)**.** *Fix N. The Taylor depth can be chosen, depending only on N, so that on $|v| \le A_N(b)$ the exact exponent may be replaced by*

$$R_{b,N}(v) = \sum_{m=1}^{2N+1} E_m(b, v) b^{-m/2}$$

with an error which, after multiplication by the Gaussian weight and integration over the central window, is

$$\mathcal{O}(b^{-N-1}).$$

The implied constant may depend on N but not on b.

Proof. All vertical derivatives of the periodic finite part are uniformly bounded because the periodic jets are smooth and bounded. Taylor's theorem therefore bounds a vertical remainder of order M by

$$C_M |\theta|^{M+1} = C_M b^{-(M+1)/2} |v|^{M+1}.$$

The logarithmic coordinate has an analogous remainder on the shrinking angular window, because $\varepsilon A_N(b) \to 0$ keeps $1 + \mathrm{i}\theta$ uniformly away from zero. Boundary jets satisfy the same kind of estimate, while the forward-product terms are exponentially small and hence smaller than any prescribed inverse power.

Choose M with enough surplus beyond $2N + 1$. On the central window, powers of $|v|$ are bounded by powers of $A_N(b)$, hence by powers of $\log b$. By lemma 11.1, the spare powers of $b^{-1/2}$ absorb every such logarithmic loss. The pointwise exponent error is therefore integrably $\mathcal{O}(b^{-N-1})$ against $\mathrm{e}^{-v^2/2}\,\mathrm{d}v$. Removing terms above order $2N + 1$ from the finite Taylor polynomial costs the same order because each carries at least the corresponding spare half-power. This leaves $R_{b,N}$. □

11.4 Finite exponentiation without an infinite-series argument

A common informal step is to exponentiate (58) term by term. The formal development instead uses a finite Taylor polynomial for the exponential and an explicit remainder.

Let

$$\mathcal{P}_N(b, v) = \sum_{\ell=0}^{2N+1} B_\ell(b, v) b^{-\ell/2}.$$

It is the reference polynomial obtained from the formal recursion in proposition 9.1.

Proposition 11.3 (Finite exponential substitution; Lean)**.** *For every fixed N,*

$$\frac{1}{\sqrt{2\pi}} \int_{|v| \le A_N(b)} \mathrm{e}^{-v^2/2} \left|\exp R_b(v) - \mathcal{P}_N(b, v)\right| \,\mathrm{d}v = \mathcal{O}(b^{-N-1}). \tag{60}$$

Proof. First replace R_b by $R_{b,N}$ using proposition 11.2. On the central window the truncated exponent tends uniformly to zero after the exact quadratic term has been removed, and it is bounded by a fixed polynomial in v times $b^{-1/2}$. Apply Taylor's formula

$$\mathrm{e}^z = \sum_{r=0}^{L-1} \frac{z^r}{r!} + \frac{z^L}{(L-1)!} \int_0^1 (1 - s)^{L-1} \mathrm{e}^{sz} \,\mathrm{d}s$$

with $L = 2(N+1)$. Uniform control of the real part of $R_{b,N}$ bounds the integral remainder by a Gaussian-integrable polynomial times b^{-N-1}.

Now expand the finite Taylor polynomial in the finitely many powers $b^{-m/2}$. The coefficient of $b^{-\ell/2}$ for $\ell < L$ is exactly B_ℓ, by the same recursion that defines the formal exponential coefficients. Algebraically, the difference between the finite exponential Taylor polynomial and $\mathcal{P}_N$ factors as

$$b^{-L/2} \times \{\text{a finite quotient polynomial in } b^{-1/2}, v, E_1, \dots, E_{2N+1}\}.$$

Because $L/2 = N+1$ and the coefficient functions are bounded, Gaussian integration gives (60). □

11.5 Gaussian contraction on the whole line

By corollary 9.2, odd ℓ contribute odd polynomials. Therefore

$$\mathfrak{G}[\mathcal{P}_N(b,\cdot)] = \sum_{j=0}^{N} a_j(b) b^{-j}. \tag{61}$$

The exact central integral, however, only runs over $|v| \le A_N(b)$. We must show that the polynomial reference can be extended to the whole real line at the required order.

Lemma 11.4 (Gaussian polynomial tail; Lean)**.** *For every fixed polynomial P and every fixed N,*

$$\int_{|v|>A_N(b)} e^{-v^2/2} |P(v)| \, dv = \mathcal{O}(b^{-N-1}).$$

The same assertion holds for a finite family of polynomials whose coefficients are uniformly bounded in b.

Proof. For every integer $q \ge 0$ there is a constant C_q such that

$$\int_A^\infty v^q e^{-v^2/2} \, dv \le C_q(1 + A^q) e^{-A^2/4} \qquad (A \ge 1).$$

At $A = A_N(b)$,

$$e^{-A_N(b)^2/4} = e^{-8(N+1)\log b} = b^{-8(N+1)}.$$

The polynomial factor in $A_N(b)$ is a power of $\log b$ and is absorbed by the enormous surplus of inverse powers. Sum over the finitely many monomials of P. □

Combining proposition 11.3 and lemma 11.4 with (61), the central contour contributes

$$\sum_{j=0}^{N} a_j(b) b^{-j} + \mathcal{O}(b^{-N-1}). \tag{62}$$

11.6 The complementary contour

The exact contour outside the central window is split into an intermediate region and a far region. The two estimates use different mechanisms.

Proposition 11.5 (All-order minor-arc bounds; Lean)**.** *For every fixed N, the normalized contour outside $|v| \le A_N(b)$ contributes $\mathcal{O}(b^{-N-1})$. More precisely:*

1. *the intermediate region has a bound with fourfold power slack, of order at most a constant times $b^{-4(N+1)}$ once the relevant dyadic index is at least $b/4$;*

2. the far region is bounded by a constant times

$$b\exp\left(-\frac{\log 2}{4}b\right),$$

which is flat against every inverse power of b.

Proof. On the intermediate region the modulus estimate for the saddle kernel contains a decay factor whose exponent is quadratic in the enlarged radius. Substitution of (59) yields

$$\exp\{-4(N+1)\log b\} = b^{-4(N+1)}$$

after the standard comparison between the dyadic index and b. The remaining geometric and polynomial factors consume far less than the three spare blocks of inverse powers.

In the far region the dyadic product supplies geometric decay in the index. Summing the geometric tail gives a bound of the displayed exponential form. For every K,

$$b^K\, b\mathrm{e}^{-(\log 2)b/4} \longrightarrow 0,$$

so this contribution is $\mathcal{O}(b^{-N-1})$ for the requested fixed N. □

11.7 The all-orders mass expansion

Theorem 11.6 (Normalized mass expansion; Lean)**.** *For every fixed $N \geq 0$, as $t \to +\infty$ with $b = \lambda(2^{-t})$,*

$$\mathcal{G}(2^{-t}) = \sum_{j=0}^{N} \frac{a_j(b)}{b^j} + \mathcal{O}(b^{-N-1}). \tag{63}$$

The coefficient functions are evaluated at the exact phase b.

Proof. The central part is (62). Add the intermediate and far parts controlled by proposition 11.5. All three errors are $\mathcal{O}(b^{-N-1})$. □

11.8 Transfer through the logarithm

Theorem 11.7 (Logarithmic expansion on the dyadic real scale; Lean)**.** *For every fixed $N \geq 0$,*

$$\log \mathcal{G}(2^{-t}) = \sum_{j=1}^{N} \frac{A_j(b)}{b^j} + \mathcal{O}(b^{-N-1}), \qquad b = \lambda(2^{-t}). \tag{64}$$

Proof. The boundedness of the a_j and (63) with $N = 0$ give $\mathcal{G}(2^{-t}) = 1 + \mathcal{O}(b^{-1})$, hence $\mathcal{G} > 0$ and $|\mathcal{G} - 1| < 1/2$ eventually. Taylor's theorem for $\log(1+z)$ is therefore uniform. Substitute (63) into the finite Taylor polynomial for the logarithm. The coefficient of b^{-j} is exactly A_j, either by (47) or by the recursion (46). The logarithmic Taylor remainder and the original mass remainder are both $\mathcal{O}(b^{-N-1})$. □

11.9 Adding the flat tail and returning to $x \downarrow 0$

The tail $\mathcal{T}$ in (56) satisfies, for every K,

$$\mathcal{T}(x) = \mathcal{O}(\lambda(x)^{-K}).$$

It therefore does not change any coefficient in a Poincaré expansion.

Theorem 11.8 (Full exact-phase expansion; Lean)**.** *For every $N \ge 0$,*

$$\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N-1}) \qquad (x \downarrow 0). \tag{65}$$

Equivalently, using a truncation with indices $j < N$,

$$\log F(x) - \mathcal{M}(x) - \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} = \mathcal{O}(\lambda(x)^{-N}).$$

Proof. First combine (56), (64), and the flatness of $\mathcal{T}$ on the scale $x = 2^{-t}$. Next use the exact change of variables $t = -\log_2 x$. As $x \downarrow 0$, $t \to +\infty$ and $\lambda(2^{-t}) = \lambda(x)$, so the dyadic-real estimate transports without approximation. This yields (65). □

Warning 11.9 (Poincaré does not mean convergent)*.* The theorem is fixed-order. For each N there is an error constant depending on N, and the central radius itself depends on N. No uniform control as $N \to \infty$ is asserted, and no convergence of the infinite coefficient series is implied.

12 A proof-oriented concordance with the Lean modules

The following table records where the two missing clusters are proved. Paths are relative to

`Analysis/FabiusFunction/Lean/FabiusFunction/`.

Table 3: Module concordance for this companion.

Module	Principal role	Status
`FabiusInverse.lean`	Order isomorphism of $[0, 1]$, totalized inverse, global continuity and monotonicity, comparison equivalences, tails, reflection, exact dyadic inversion, transported flatness, effective steepness, and endpoint quotient divergence.	Lean
`FabiusSaddleJet` `ClosedForm.lean`	Operator solution of the recursive periodic jets and the harmonic-number constant term.	Lean
`FabiusSaddleJet` `Stirling.lean`	Identification of the operator coefficients with $s(n + 1, m + 1)$ and the corresponding closed derivative formula.	Lean
`FabiusSaddleExponent` `ClosedForm.lean`	Collapse of each centered exponent coefficient to one bounded jet monomial plus one universal monomial; parity.	Lean
`FabiusSaddleExpansion` `Coefficients.lean`	Formal exponential coefficients, Gaussian contraction, mass coefficients, logarithmic transfer, periodicity, smoothness, and boundedness.	Lean
`FabiusSaddleLeading` `Coefficient.lean`	Affine dependence of a_j on the newest jet and coefficient $(-1)^j/(2^j j!)$.	Lean
`FabiusSecondSaddle`		

Module	Principal role	Status
`Correction.lean`	Order-four exponential identity, explicit B_4, Gaussian contraction, a_2, A_2, and regularity of the second correction.	Lean
`FabiusSecondSaddle Expansion.lean`	Insertion of A_2 into the exact-phase asymptotic with a third-order remainder.	Lean
`FabiusSaddleCentral AllOrders.lean`	Exact central decomposition, finite reference polynomial, odd-term cancellation, and integrated central remainders.	Lean
`FabiusSaddleTail AllOrders.lean`	Order-dependent radius and complementary intermediate/far contour bounds of arbitrary fixed inverse-power order.	Lean
`FabiusSaddleMass AllOrders.lean`	Assembly of central Taylor control, finite exponential substitution, whole-line Gaussian contraction, and minor arcs into the complete mass expansion.	Lean
`FabiusFullAsymptotic Expansion.lean`	Transfer from mass to logarithm, addition of the flat forward tail, and transport from $x = 2^{-t}$ to the full limit $x \downarrow 0$.	Lean

12.1 Named declarations worth knowing

The inverse interface is centered around

```
fabiusOrderIso
fabiusReal_fabiusInv
fabiusInv_fabiusReal
strictMonoOn_fabiusInv
monotone_fabiusInv
continuous_fabiusInv
fabiusInv_le_iff_le_fabiusReal
fabiusReal_le_iff_le_fabiusInv
fabiusInv_lt_iff_lt_fabiusReal
fabiusReal_lt_iff_lt_fabiusInv
fabiusInv_one_sub
fabiusInv_fabiusDyadicUnit
le_two_pow_mul_fabiusInv_pow
le_two_pow_div_factorial_mul_fabiusInv_pow
mul_lt_fabiusInv
tendsto_fabiusInv_div_atTop
tendsto_one_sub_fabiusInv_div_one_sub_atTop
```

The all-orders endpoint interface culminates in

```
fabiusSaddleKernelMass_dyadicLambert_hasAsymptoticExpansion
fabiusSaddleRatio_dyadicLambert_hasAsymptoticExpansion
log_fabius_dyadicReal_sub_sharpLambertMain_hasAsymptoticExpansion
log_fabius_sub_sharpLambertMain_hasAsymptoticExpansion
log_fabius_sub_sharpLambertExpansion_isBigO
```

These names are not needed to read the mathematics, but they make the article useful as a map back into the source tree.

13 What is formalized and what is newly synthesized

For clarity, the principal statements can be divided into three groups.

13.1 Directly formalized

The following content is represented directly by declarations in the repository:

- the global totalized inverse package and all four comparison equivalences;
- the constant tails, reflection, midpoint, and exact dyadic inversion;
- the effective and sharp inverse power inequalities, effective linear steepness, and the two endpoint quotient limits;
- the closed jet formulas, shifted Stirling conversion, and closed exponent coefficients;
- the formal exponential and logarithmic coefficient machinery, top-jet coefficient, first and second corrections;
- the arbitrary-order central, tail, mass, and logarithmic asymptotic theorems.

13.2 Classical consequences derived in this article

The following statements are proved here from the formalized input:

- C^∞ interior regularity of G and the formulas for G' and G'';
- strict concavity on $(0, 1/2)$, strict convexity on $(1/2, 1)$, and the exact smooth and analytic loci;
- divergence of $G(y)/y^\alpha$ for every $\alpha > 0$;
- the sharp equivalent (19) and the scale hierarchy (22)–(23);
- the explicit all-j composition sum (44) and the highest-Ψ-derivative corollary (49).

13.3 Claims deliberately not made

This companion does *not* claim:

- convergence or Borel summability of the all-orders series;
- an all-orders substitution of an approximate Lambert phase inside the periodic coefficients;
- a closed form for the inverse at arbitrary rational arguments;
- analyticity of either F or G inside the transition interval;
- uniformity of the remainder constants as the truncation order tends to infinity.

14 Concluding synthesis

The inverse function and the saddle coefficient machinery illuminate opposite sides of the same endpoint phenomenon.

On the geometric side, F approaches zero more rapidly than every power. Its inverse must therefore leave zero more rapidly than every power, and the formalized comparison calculus turns this slogan into effective inequalities with exact constants. The sharp Lambert expansion goes further: the inverse lives on the stretched-logarithmic scale

$$G(y) \asymp \sqrt{\log(1/y)}\,\exp\Bigl(-\sqrt{2\log 2\ \log(1/y)}\Bigr),$$

with the exact leading constant $1/(\mathrm{e}\sqrt{\log 2})$. Reflection transfers the entire picture to the endpoint 1.

On the asymptotic side, the apparent complexity of the periodic corrections is organized by three simple structures. First, the recursive jets are solved by a product of commuting first-order operators; the forcing terms sum to a harmonic number and the derivative coefficients are shifted signed Stirling numbers. Second, centering collapses every exponent coefficient to two monomials, forcing parity and eliminating all odd half-powers under Gaussian contraction. Third, the newest jet can enter only linearly, so the coefficient of the highest derivative $\Psi^{(2j)}$ is known at every order before the lower-order algebra is performed.

The analytic remainder proof then explains why these formal coefficients are not merely symbolic. The growing window $A_N(b) = \sqrt{32(N+1)\log b}$ separates the local and global tasks: locally it is still narrow enough for uniform Taylor expansion, while globally it is wide enough to make both Gaussian and contour tails smaller than the requested inverse power. Finite exponential substitution and logarithmic transfer complete a genuine Poincaré expansion at the exact Lambert phase.

Together, these missing parts close two expository loops. Flatness of F is now matched by a precise theory of the steep inverse, and the headline all-orders expansion is now matched by a transparent account of the coefficient algebra and the analytic estimates that make it true.

A A compact inverse reference

For convenience, here is the inverse theory in one place. For $G(y) = F_I^{-1}(\mathrm{clamp}_{[0,1]}(y))$:

$$\begin{aligned}
&0 \le G(y) \le 1, \qquad G(y) = 0\ (y \le 0), \qquad G(y) = 1\ (y \ge 1),\\
&F(G(y)) = y \quad (0 \le y \le 1), \qquad G(F(x)) = x \quad (0 \le x \le 1),\\
&G(1-y) = 1 - G(y), \qquad G(1/2) = 1/2,\\
&G(y) \le x \iff y \le F(x), \qquad F(x) \le y \iff x \le G(y),\\
&G(y) < x \iff y < F(x), \qquad F(x) < y \iff x < G(y),\\
&G(\mathsf{F}_n(a)) = \min\{a, 2^n\}/2^n,\\
&y \le C_n G(y)^n, \qquad y \le C_n G(y)^n/n! \quad \text{when } 2^n G(y) \le 1,\\
&G(y)/y \to +\infty \quad (y \downarrow 0), \qquad (1-G(y))/(1-y) \to +\infty \quad (y \uparrow 1),\\
&G(y) \sim \frac{\sqrt{\log(1/y)}}{\mathrm{e}\sqrt{\log 2}}\,\exp\Bigl[-\sqrt{2\log 2\ \log(1/y)}\Bigr].
\end{aligned}$$

B A compact coefficient reference

With $L = \log 2$,

$$
\begin{aligned}
p_0 &= \frac{1}{2} + \frac{\Psi'}{L},\\
p_{n+1} &= (D/L - (n+1))p_n + \frac{(-1)^{n+1} n!}{L},\\
d_n &= p_n + (-1)^{n+1} n!,\\
c(n,m) &= s(n+1, m+1),\\
d_n &= (-1)^n n! \left(\frac{H_n}{L} - \frac{1}{2}\right) + \sum_{m=0}^{n} c(n,m) \frac{\Psi^{(m+1)}}{L^{m+1}},\\
E_m &= \mathrm{i}^m \left(\frac{d_{m-1}}{m!} v^m + \frac{(-1)^{m+1}}{m+2} v^{m+2}\right),\\
nB_n &= \sum_{m=1}^{n} m E_m B_{n-m},\\
a_j &= \mathfrak{G}[B_{2j}],\\
A_n &= a_n - \frac{1}{n} \sum_{k=1}^{n-1} k A_k a_{n-k},\\
[d_{2j-1}]\, A_j &= \frac{(-1)^j}{2^j j!},\\
[\Psi^{(2j)}]\, A_j &= \frac{(-1)^j}{2^j j! L^{2j}}.
\end{aligned}
$$

The first two logarithmic coefficients are

$$
\begin{aligned}
A_1 &= -\frac{d_0^2}{2} - d_0 - \frac{d_1}{2} - \frac{1}{12},\\
A_2 &= \frac{1}{24}\big(8d_0^3 + 12d_0^2 d_1 + 12d_0^2 + 48 d_0 d_1 + 12 d_0 d_2\\
&\qquad + 6d_1^2 + 24 d_1 + 20 d_2 + 3 d_3\big).
\end{aligned}
$$

C Repository paths used in the audit

```
Analysis/FabiusFunction/
  Lean/FabiusFunction/FabiusInverse.lean
  Lean/FabiusFunction/FabiusSaddleJetClosedForm.lean
  Lean/FabiusFunction/FabiusSaddleJetStirling.lean
  Lean/FabiusFunction/FabiusSaddleExponentClosedForm.lean
  Lean/FabiusFunction/FabiusSaddleLeadingCoefficient.lean
  Lean/FabiusFunction/FabiusSecondSaddleCorrection.lean
  Lean/FabiusFunction/FabiusSecondSaddleExpansion.lean
  Lean/FabiusFunction/FabiusSaddleCentralAllOrders.lean
  Lean/FabiusFunction/FabiusSaddleTailAllOrders.lean
  Lean/FabiusFunction/FabiusSaddleMassAllOrders.lean
  Lean/FabiusFunction/FabiusFullAsymptoticExpansion.lean
  docs/Fabius_Function_and_Rvachev_Up/
    Fabius_Function_and_Rvachev_Up.tex
```

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